Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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Abstract

A decoder fit to a singular animal's cortex drifts, losing accuracy over days. Usually the remedy is recalibrating on fresh data from the same subject. We report a different kind of robustness: a decoder built from one other individual, or pooled from a set of other individuals, ages more slowly than the subject's own. On the open BraiDyn-BC cohort (25 mice, 44 Allen-atlas cortical parcels, a $\sim$15-day operant protocol) we decode four task events and measure, per mouse, the AUC lost over two weeks by its own early-week decoder against one pooled over the other 24 mice and one built from a single other mouse. The personal decoder ages more by $+0.017$ AUC ($p = 5 \times 10^{-5}$, 73 of 100 mouse–event pairs). The delta from each effect varies among them, but they are chained within each trial – every reward onset has a lick within $0.12\,\mathrm{s}$ – and they differ highly in sample size, so we report the variation in general without attributing it to the individual behaviors. This also is not a data-volume effect: at a matched training-event count a decoder from a single other mouse still ages less. Hence we conclude this is rather an effect of not overfitting an individual's drift-prone idiosyncrasies. It also survives a 1-D CNN and a GRU. The early-week accuracy cost of decoding across individuals is repaid as late-week stability, which suggests population priors as an alternative to per-subject recalibration. Code and a streamed-feature pipeline are released.

1 Introduction

Representational drift, the reorganization of an animal's neural code over days with fixed behavior, is now documented across sensory, motor, and association cortex and in the hippocampus [17, 2, 19]. This has a stark impact on neural decoding: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is recalibration, periodically refitting the decoder on fresh within-subject data [1, 4], which requires ongoing labelled data from the individual and treats each subject in isolation.

The code's cross-individual structure offers an alternate route to temporal robustness. Cortex-wide behavioral representations are considerably shared among individuals [10, 6, 7], which already allows models to decode behavior from unseen individuals [20]. The question we pose is temporal: does a decoder built from a population of other individuals resist drift better than a decoder built from the subject itself? This prediction, to our knowledge, has not been examined.

We test it on BraiDyn-BC [5], a widefield calcium-imaging resource whose $\sim$15-day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time. Our contributions:

- **Pooling resists drift.** Across a two-week period the population-pooled decoder loses less accuracy than the individual's own, with a paired asymmetry of $+0.017$ AUC ($p = 5 \times$

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10^{-5}, 73 of 100 pairs; Section 4.1). The size of this effect varies among the four events, but those are chained within a trial and have highly differing counts, so we treat the pooled estimate as the result and leave the per-event ranking uninterpreted (Section 4.2).

- **It is not a data-volume artifact.** At a matched training-event count a decoder trained on one other mouse ages less than one trained on the subject's own, by $+0.010$ AUC on average. The gain therefore comes from training on individuals other than the subject, not from the larger training set pooling usually brings, and training on many other mice rather than one lowers aging further on all four events (Sections 4.4–4.5).
- **Session-level validation is load-bearing.** As trials within a recording session are dependent, a trial-level split inflates the personal decoder but not the pooled one, one-sidedly inflating the asymmetry. We report that inflation for each event; on lever pull it accounts for the entire apparent effect. For longitudinal decoding comparisons we recommend session-held-out validation as a default (Section 4.6).

2 Related work

Representational drift and its mitigation. Drift is characterized within individual subjects [17, 2, 19, 15], and most approaches to mitigating it also operate within the subject. These include manifold stabilization [1], supervised and self-calibrating recalibration [4, 14], and adversarial or nonlinear alignment to the subject's own reference day [3, 13]. Each method uses later data from the same subject to stabilize the decoder. We instead ask whether data from other subjects can provide that stability.

Cross-subject decoding. Shared structure across individuals already supports cross-subject readout through manifold alignment [8], contrastive embeddings [18], and, on this cohort, atlas-aligned foundation models [20]. Multi-participant pooling can also reduce overfitting: HTNet, for example, pools participant data to avoid fitting a decoder to a small single-subject sample [12], and pooled pretraining across 83 animals followed by fine-tuning on held-out animals reaches state-of-the-art encoding and decoding on a standardized cross-subject cohort [11]. Our question differs in two ways. Primarily, prior work measures accuracy or transfer at a fixed time, while we look at how that accuracy changes over days, to see whether pooled decoders age more slowly. Further, we keep the training-set size fixed with the count-matched control, hence being able to delegate the relevant overfitting not to limited data but to idiosyncratic features that cause drift.

The closest work to ours combines both axes: Meta-AlignNN [9] uses the consistency of neural population dynamics across individuals to build a decoder that remains stable across subjects, time and tasks, and demonstrates it over two years in three monkeys. That work and ours point in the same practical direction, and differ in kind rather than in topic. Meta-AlignNN proposes a method for achieving stability; we measure whether a difference in aging rate exists at all between a decoder trained on the subject and one trained on others, with the training-set size held fixed so that the comparison isolates whose data the decoder saw. A measured effect of this kind is what would justify the design choice such methods already make.

Conservation of cortex-wide dynamics. Cortex-wide activity contains shared, low-dimensional, movement-related components [10, 6, 7], and latent dynamics are preserved across animals performing similar behaviour to the point that a decoder fit to one animal transfers to another [16]. That conservation is the mechanism a pooled decoder would exploit, and it is why pooling is possible at all here. It does not, however, imply that pooling will *resist drift*: a code can be shared across individuals and still move within each of them at the same rate. Whether the shared component is also the more temporally stable one is the empirical question we ask.

3 Data and methods

Dataset. BraiDyn-BC [5] (DANDI:001425, CC-BY) provides, for each mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ traces parcellated into 44 Allen Common Coordinate Framework regions at $\sim$30 Hz, together with time-aligned behavioral channels. The dataset includes 25 mice, each of which completed $\sim$15 daily operant sessions over $\sim$3 weeks. We stream the 44-region traces and behavioral channels without downloading the raw movies.

Features and events. We detect the onset of four events (lever pull, lick, reward, and tone) as rising edges, with a 1 s refractory period. For each onset we construct a 44-dimensional evoked feature by subtracting the mean pre-onset baseline (-0.5–0 s) from the mean post-onset $\Delta F/F$ response (0–1 s) in each parcel. Negative samples are drawn from matched-count windows more than 2 s from any onset *of the same event*; they are not screened against the other three events, and Section 4.2 reports how often they contain one. Every classification is a binary decode of one event against these matched negatives, scored by area under the ROC curve (AUC). The four decoders are therefore four independent binary detectors, not a four-way classifier: nothing in this paper asks a model to distinguish one event from another.

Decoders. The primary decoder is a standardized ℓ_2-regularized logistic regression on the 44-dimensional evoked feature. This is a linear readout of the parcel-space code: it recovers the component of behavior that is linearly present in the population average, and it is the natural comparison to prior cross-subject decoders, which are overwhelmingly linear [8, 12]. Every condition below uses this same readout, so the only variable is which data the decoder was trained on.[1]

Temporal blocks and aging. We group each mouse's sessions into an early block (days 1–5) and a late block (days 11–15) and report AUC under four conditions. The personal decoder is estimated by leaving out one whole early *session* at a time: for each early session s we train on the mouse's remaining early sessions and score the resulting decoder both on s (within-mouse same-block, WS) and on the late block (within-mouse across-block, WX), then average over s. Because WS and WX come from the same fitted decoders, they differ only in which data they are tested on. The pooled decoder is trained once on the other mice's early blocks and scored on the same held-out sessions s (LS) and on the late block (LX).

Holding out sessions rather than trials is essential and not a detail. Trials within one recording share illumination and hemodynamic state, slow arousal drift, and temporal autocorrelation, so a trial-level split lets the personal decoder train and test inside the same session. The pooled decoder is trained on other animals entirely and cannot benefit this way, so the inflation is one-sided and lands directly on the quantity of interest; Section 4.6 reports what it costs per event.

Aging is the accuracy lost when testing moves from the held-out mouse's early data to its late block while training stays fixed: personal aging is $\mathrm{WS} - \mathrm{WX}$ and pooled aging is $\mathrm{LS} - \mathrm{LX}$. The per-mouse aging asymmetry is $(\mathrm{WS} - \mathrm{WX}) - (\mathrm{LS} - \mathrm{LX})$; a positive value means the personal decoder ages more. We assess significance with a paired sign-flip permutation test across mice (20,000 permutations) and compute 95% confidence intervals with a cluster bootstrap over mice.

4 Results

4.1 A population-pooled decoder ages more slowly than a personal one

For each mouse we keep the test data fixed (first the mouse's held-out early sessions, then its late block) and change only the source of the training data: either the mouse itself or the other 24 mice. Across all 100 mouse–event pairs the asymmetry is $+0.017$ AUC ($p = 5 \times 10^{-5}$), with the personal decoder aging more in 73% of pairs (Table 1).

The effect differs among the four events: it is largest for reward ($+0.029$, $p = 0.0005$) and lick ($+0.020$, $p = 0.002$), marginal for tone ($+0.015$, $p = 0.08$), and absent for lever pull ($+0.003$, $p = 0.37$). While we report this variation, we do not interpret it, as the four events all fall within roughly the same two seconds of a trial (Section 4.2). The pooled estimate is the claim we defend; the per-event column is data.

[1] The result is not a property of the linear readout. Repeating the analysis with a 1-D convolutional network over time and a GRU over the frame sequence (full evoked window, -0.5 to $+1.0$ s, 45×44), under the same session-held-out protocol, the asymmetry survives for both events that carry it: lick gives $+0.020$, $+0.012$ and $+0.010$ and reward $+0.029$, $+0.015$ and $+0.012$ under the linear, CNN and GRU readouts, all significant. Under a trial-level split the nonlinear estimates appear *larger* instead, which is the inflation of Section 4.6: a higher-capacity model exploits it more effectively than a linear one.

Table 1: Personal vs. pooled aging across days (AUC lost from early to late test, holding training fixed), per event, $n = 25$ mice, with whole early sessions held out. Asymmetry = personal − pooled aging; a positive value means the mouse's own decoder ages more. p: paired sign-flip permutation. The rightmost column gives the same asymmetry under a trial-level split, the protocol this analysis replaces (Section 4.6).

Event	Personal	Pooled	Asymmetry (95% CI)	p	own ages more	trial-split
Lever-pull	-0.022	-0.025	$+0.003$ $[-0.004, 0.011]$	0.37	64%	$+0.014$
Lick	$+0.012$	-0.009	$+0.020$ $[0.008, 0.032]$	0.0023	80%	$+0.028$
Reward	$+0.017$	-0.012	$+0.029$ $[0.014, 0.046]$	0.0005	80%	$+0.032$
Tone	$+0.016$	$+0.001$	$+0.015$ $[0.001, 0.032]$	0.077	68%	$+0.022$
Pooled			$\mathbf{+0.017}$ $[0.010, 0.024]$	$\mathbf{5 \times 10^{-5}}$	**73%**	$+0.024$

4.2 What the four events are

The four decoded events come from a single cued lever-pull trial structure, so they are chained in time rather than independent. Every reward onset has a lever-pull onset a median of 0.083 s away and a lick onset 0.117 s away, and 100% and 98% respectively fall inside the window from which a feature vector is built. Across all four events, the fraction of positive windows containing at least one other event is 100% (reward), 84% (lick), 84% (tone) and 68% (lever pull). The reward decoder is therefore never reading reward in isolation; its input window always contains licking and lever movement as well. The negatives are not event-free either: screened only against the event's own onsets, they contain another event 33–57% of the time.

The events also differ heavily in sample size, from a median of 56 reward onsets per session against 107 (tone), 530 (lick) and 652 (lever pull). The per-event asymmetry runs roughly opposite to that count; a decoder fit to fewer events has more idiosyncrasy available to overfit, and hence more for pooling to avoid. We cannot separate that explanation from a behavioral one with four events, and we do not try. This is why the pooled estimate over all 100 pairs is what we defend, and why later sections that select reward and lick are consistency checks on the same events rather than evidence from independent behaviors.

4.3 The personal decoder's advantage erodes over the protocol

The block split compares two windows, but the effect should also appear as a continuous trend across all fifteen days. For each session day we compute the accuracy of the personal decoder and the pooled decoder and regress each on day; on early days the personal decoder is trained by leave-one-session-out, so it is never tested on a session it trained on. The personal decoder starts ahead, leading by about 0.035 AUC on day 1 as expected for a decoder trained on the same animal. That lead then erodes: the gap narrows by 0.0014 AUC per day ($p = 5 \times 10^{-5}$; narrowing in 65% of pairs), so by day 15 roughly 60% of the early-week advantage is gone.

Averaged over the four events, the pooled decoder gains accuracy over days (+0.0015 AUC/day) as the sessions become easier – the mouse is more practiced – while the personal decoder stays flat (+0.0000 AUC/day). The personal decoder, fit to idiosyncrasies, is unable to ride that wave, and the lift it fails to take is what opens the gap. Per event, the personal decoder does lose accuracy for lick (−0.0010 AUC/day) and reward (−0.0016 AUC/day), and these are the two events where the narrowing is individually significant ($p = 0.007$ and $p = 0.001$; Fig. 1).

4.4 The robustness is not a data-volume artifact

A pooled decoder contains $\sim$24× more training events than a personal decoder, so its stability could reflect data volume rather than pooling. We test this with a count-matched control (Fig. 2, left). For each held-out early session s of each mouse we set the training-set size to n, the number of events in that mouse's remaining early sessions, and compare three decoders: *self*, trained on those n events; *one*, trained on n events from a single other mouse; and *many*, trained on n events drawn across the other 24 mice. All three are scored on the same held-out session s and the same late block, so they differ only in whose data they were fit on.

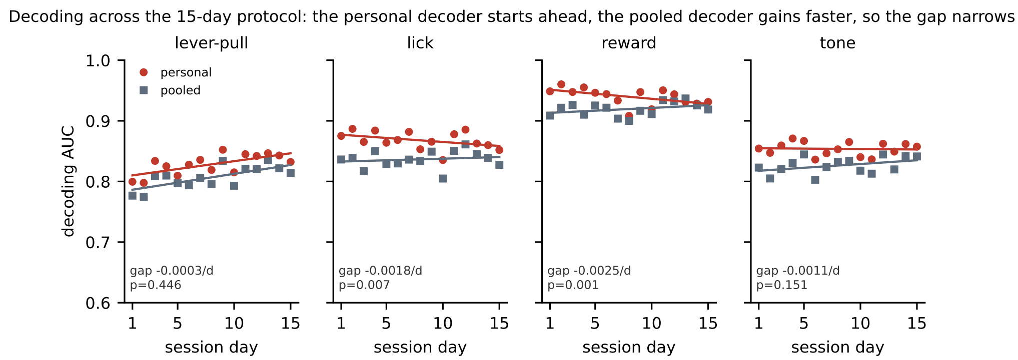

Figure 1: Decoding accuracy across the 15-day protocol for the personal decoder (own early data) and the pooled decoder (other mice's early data), with per-day points and linear fits. The personal decoder starts ahead, and the pooled decoder gains more over days, so the gap between them narrows. The narrowing is significant for lick and reward, the two events where the personal decoder also loses accuracy over days, and is not significant for lever pull or tone. The per-event gap slope and its p (paired sign-flip over mice) are shown.

For the events where there was an asymmetry, the count-matched ordering reproduces it. The self decoder ages more than a decoder trained on one other mouse by $+0.010$ AUC on average, and by $+0.007$ for lick, $+0.017$ for reward and $+0.006$ for tone individually. As both decoders see the same number of training events, this is not a data-volume artifact.

The diversity ordering is the most robust result in this analysis: *many* ages less than *one* for all four events, including lever pull (-0.026 vs. -0.020). Training on a diverse set of individuals rather than a single other individual therefore provides a smaller, additional benefit that does not depend on which event is decoded.

4.5 Diversity: a secondary scaling with the number of training individuals

We next vary the number of training mice, $N \in \{1, 2, 4, 8, 16, 24\}$ (Fig. 2, right). As N increases, pooled aging becomes slightly more favorable for lick (-0.001 to -0.010), reward (-0.004 to -0.010), and tone ($+0.003$ to -0.010); lever pull saturates with a single training mouse and remains near -0.025. The scaling effect is around 0.01 AUC, comparable to the self-versus-one-other gap of Section 4.4. Most of the resistance to drift is already present when training on a single other individual, and additional individuals refine it further. This analysis never lets the decoder see the held-out mouse, so it is unaffected by the protocol issue below.

4.6 How much the validation protocol matters

Every number above holds out whole sessions. Repeating the identical analysis on the identical features with a trial-level split – shuffling trials pooled across the early sessions, so that a decoder may train on one part of a session and be tested on another – inflates the personal decoder's within-block accuracy by $+0.013$ (lever pull), $+0.010$ (lick), $+0.009$ (reward) and $+0.000$ AUC (tone). The pooled decoder is unaffected, because it is trained on other animals and has no session in common with the test data.

The inflation therefore lands on one side of a difference between two arms (rightmost column of Table 1). For lever pull it is the entire effect: a significant-looking $+0.014$ ($p = 0.001$) becomes $+0.003$ ($p = 0.37$). For reward, which has the largest true effect, it changes little. The same reasoning applies to the count-matched control, where the natural implementation is to score the self decoder on the block it was fit on – not a cross-validated quantity at all, and enough to reverse the ordering the control exists to establish.

We report this because the pattern generalizes beyond our dataset. Any comparison of a within-subject decoder against a cross-subject one is exposed to it, since only the within-subject arm can

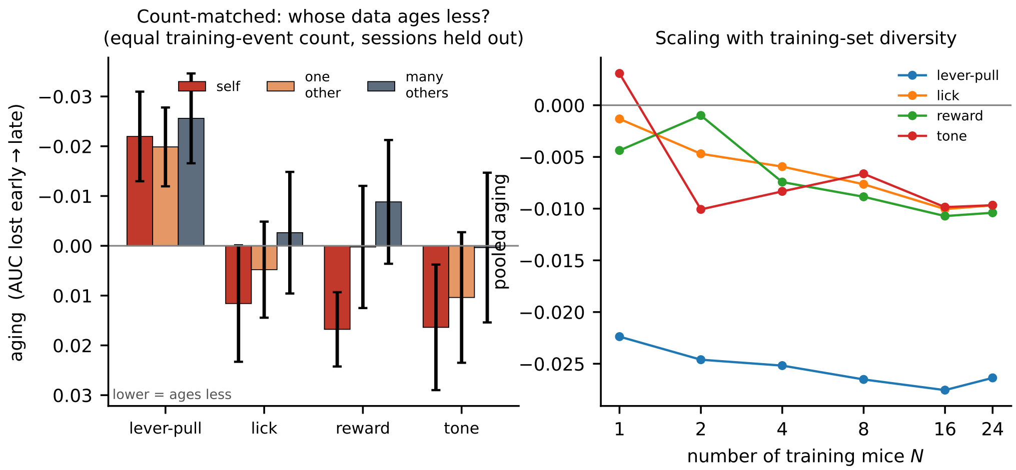

Figure 2: **Left:** count-matched control. With equal numbers of training events, decoder aging (AUC lost from early to late; lower values indicate greater robustness) is compared for the subject's own data (self), one other mouse, and many other mice. **Right:** pooled aging as a function of the number of training mice N. Greater diversity provides a small additional gain, with immediate saturation for lever pull.

share a session between train and test. Where trials are not independent – which for physiological recordings is the normal case – the split must respect the recording session.

5 Discussion

A subject's own decoder overfits idiosyncratic features of its early sessions, some of which later drift. A decoder that has never seen the subject must instead rely on components of the code that are shared across individuals and more stable over time. The count-matched control supports this reading directly: at equal data, other-individual data ages less than the subject's own. Population structure is therefore not only a resource for cross-subject transfer but also for temporal robustness, and the early-week accuracy cost of cross-subject decoding is recovered over time as the pooled decoder remains stable. Unlike within-subject recalibration, this requires no newly labelled data from the individual at test time – which is the practical point, since obtaining that data is the expensive part of deploying a decoder over months.

The effect is small in absolute terms, and we think it is worth being explicit about why we believe it anyway. It rests on 25 animals and 357 recording sessions, which is a large cohort for a longitudinal cortex-wide study, and it is supported by four checks that are not variations on one analysis. A two-block contrast and a day-by-day regression over all fifteen sessions are different estimators and select the same events. The count-matched control fixes the training-set size, so the result cannot be a data-volume artifact. Replacing the linear readout with a 1-D CNN or a GRU leaves it in place, so it is not an artifact of a weak decoder. And holding out whole sessions rather than trials removes an inflation large enough to eliminate one of the four events entirely – the effect reported here is what remains after that correction, not before it. Each of these could have removed the result and none did.

Limitations. The clearest limitation is what the four events are. They are not four independent behaviors but four thresholded channels of one cued lever-pull trial, overlapping to the point where 100% of reward windows also contain licking, and differing tenfold in event count. We therefore cannot always say which behavior the effect belongs to, nor can we exclude the idea that the per-event ranking is a tracker of how much data each decoder had to overfit. A design with temporally separated, count-matched events would settle both, and is the experiment we would run next.

We also measure at the mesoscale, using parcel-averaged activity, so whether the same result holds for single-neuron codes – where representational drift is best characterized – remains unknown, and the analysis covers a two-week interval in one species. Finally, the trial-level inflation of Section 4.6 is worth stating as a caution as much as a result: it is large enough to manufacture an effect that is not there, and any comparison of this shape should be reported with sessions held out.

Reproducibility

BraiDyn-BC is publicly available through DANDI:001425. We release the analysis code and a streaming feature pipeline that reproduces all reported results from the parcellated traces without downloading the raw imaging data. Section 3 and the released code specify all data splits, permutation seeds, and hyperparameters.

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